Please Review Your Abstract. If The Abstract Is Final, Click "Submit Abstract" To Complete Your Submission.

Submit Abstract

Statistical Models for Tropical Cyclone Prediction in the North Atlantic Ocean through Geographic Sub-Basin Identification

Allison Clancy, University of California Davis, Davis, CA, United States, Paul Aaron Ullrich, Lawrence Livermore National Laboratory, Livermore, United States; University of California Davis, Davis, United States and Yushan Han, University of California Davis, Department of Land, Air and Water Resources, Davis, CA, United States

Abstract Text:
The number and character of North Atlantic Ocean tropical cyclones (TCs) vary significantly each year, driven in part by the relative favorability of large-scale environmental conditions. However, because of a limited historical record, the strength of this response across geographic regions within the Atlantic basin is not well understood. Here we develop geographic sub-basins within the North Atlantic by performing a cluster analysis for key TC genesis factors such as sea surface temperature (SST), genesis potential index (GPI), and vertical wind shear and manually adjusting boundaries to capture gradients in TC origin location and density. For each sub-basin, a regression model is developed in order to evaluate the importance of variables that contribute to TC genesis and sustainability in comparison to other sub-basins. These models estimate frequency, lifespan, and intensity of TCs in a specific sub-basin, as well as evaluate their response to key oceanic modes of variability. Statistical testing is performed to determine the significance of each predictor within a sub-basin and the distinctiveness across sub-basins. From this set of statistical models, we can evaluate how commonly used thresholds for TC development (SST, GPI, shear) vary in magnitude and importance across sub-basins. Overall, this approach allows us to better organize and categorize tropical cyclones and better tease apart the factors that drive their emergence and evolution in this region.

Session Selection:
A119. Tropical Cyclones: Observations, Modeling and Predictability - Today and into the Future

Submitter's E-mail Address:

aeclancy@ucdavis.edu

Abstract Title:
Statistical Models for Tropical Cyclone Prediction in the North Atlantic Ocean through Geographic Sub-Basin Identification

Requested Presentation Type:
Assigned by Program Committee (oral, poster, or GeoBurst)

Recording Permission Given?
Yes

Previously Published?:
No

Abstract Payment:
Paid (agu-agu26-2091346-7000-5679-1040-2750)

Check off to signify you read the statement above.

First Presenting Author
Presenting Author

Allison Clancy
Primary Email: aeclancy@ucdavis.edu

Affiliation(s):

University of California Davis
Davis CA 95616 (United States)

University of California Davis
Davis CA 95616 (United States)

Second Author

Paul Aaron Ullrich
Primary Email: ullrich4@llnl.gov

Affiliation(s):

University of California Davis
Davis 95616 (United States)

Lawrence Livermore National Laboratory
Livermore (United States)

Third Author

Yushan Han
Primary Email: yshhan@ucdavis.edu

Affiliation(s):

University of California Davis
Department of Land, Air and Water Resources
Davis CA 95616 (United States)

FINAL STEPS

1. **Check spelling and contact information.**
2. **Make necessary corrections:**
   - Select the step in the Abstract Control Panel that you wish to edit (e.g., Authors, Abstract Details)
   - Edit the information and click the submit button.
3. **Click** here **to print this page now.**